White Paper

# Right-Sizing Signal Detection: Matching Pharmacovigilance Methodology to Organizational Scale

*Why fit-for-purpose signal management, anchored by expert judgment and human-in-the-loop oversight, outperforms one-size-fits-all approaches*

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## Introduction

Signal detection protects patients. Through it, organizations identify, evaluate, and manage emerging safety risks across a product's entire lifecycle, and the regulatory objective never wavers: surface meaningful safety signals early, assess them rigorously, and act on them with transparency and accountability. How organizations meet that objective is another matter entirely.

A small biotech monitoring a single investigational asset does not operate like a global pharmaceutical company managing hundreds of products across dozens of markets. The data volumes differ by orders of magnitude. Governance structures reflect different levels of maturity, and technology capabilities are rarely comparable. Yet many organizations select a signal detection methodology by default rather than by design, inheriting a process that fits a different stage of growth than the one they occupy.

The evidence settles the question. No single approach suits every organization. Effective signal detection depends on a fit-for-purpose model that balances operational efficiency, scientific rigor, and regulatory compliance. As an organization grows, its methods evolve from manual, medically driven review toward analytics-enabled approaches supported by advanced technology and structured governance. One element holds constant throughout that evolution: expert medical judgment.

## The regulatory constant and the operational variable

Regulatory expectations for signal management are consistent across the industry. Health authorities expect timely identification of safety signals, sound scientific evaluation, documented decisions, and demonstrable inspection readiness. These expectations do not scale down for smaller companies or relax for larger ones.

The operating reality behind those expectations is where organizations diverge. Safety data volume, portfolio complexity, governance maturity, and technology capability all shape execution. Defining organizational size by these functional characteristics, rather than by revenue, is more useful. What drives a company's signal detection needs is how much safety data it processes, how complex its portfolio is, and how mature its governance has become.

Seen through that lens, three models emerge along the maturity spectrum: clinical-led, hybrid, and analytics-enabled. Each fits a distinct stage of organizational growth.

*Neither EU GVP Module IX, ICH E2E, nor FDA Good Pharmacovigilance Practices prescribe a uniform signal detection methodology for all organizations. Instead, these frameworks consistently support a risk-based, scientifically justified, and proportionate approach in which signal detection activities are aligned with product risk, data volume, lifecycle stage, and organizational capabilities.*



# How signal detection evolves across organizational scale

The comparison below highlights how signal detection methodologies evolve as organizations expand in scale and complexity.

| DIMENSION | SMALL PHARMA/BIOTECH | MID-SIZE PHARMA | LARGE PHARMA |
|---|---|---|---|
| **Data volume and complexity** | Low, focused datasets | Moderate, expanding datasets | High-volume, global datasets |
| **Primary methodology** | Qualitative, case-by-case medical review | Hybrid qualitative and quantitative review | Multilayered, analytics-driven approach |
| **Data sources** | Internal ICSRs, Literature, Health Authority (HA) communications | Internal ICSRs, Literature, HA, EudraVigilance | Internal and external sources (AERS, VigiBase, Real World Data) |
| **Statistical tools** | Spreadsheets, Frequency review | PFRR (Product Fractional Reporting Ratio), PRR (Proportional Reporting Ratio), ROR (Reporting Odds Ratio), trend analyses | EBGM (Empirical Bayes Geometric Mean), IC (Information Component), advanced analytics |
| **Signal identification focus** | Serious and unexpected adverse events | Serious events and prioritized trends | Early detection across stratified populations |
| **Analytical techniques** | Medical judgment, biological plausibility | Disproportionality and trend reviews | Automated screening and advanced stratification |
| **Governance** | Lean and informal oversight | Formal signal review committees | Global multi-layer governance frameworks |
| **Technology enablement** | Primarily manual processes | Semi-automated dashboards and workflows | Enterprise signal detection platforms |
| **Role of human expertise** | Primary decision-maker | Decision-maker supported by analytics | Oversight and validation of analytical outputs |

## Small pharmaceutical and biotechnology organizations: A clinical-led model

With low, focused datasets, small organizations can put a physician at the center of every review. Medical judgment is the analytical engine, and clinical experts assess biological plausibility and causality case by case, staying close to the underlying data. The human expert is the primary decision-maker.

The approach delivers strong scientific assessment and direct oversight. It does not scale. As case volumes climb, a process built on manual, individualized review consumes disproportionate resources and eventually strains under its own weight. Companies that recognize this ceiling early plan their next stage before volume overtakes capacity.

## Mid-size pharmaceutical organizations: A hybrid model

As portfolios expand and datasets reach moderate size, signal detection becomes more structured. Quantitative screening now runs alongside qualitative medical assessment, and formal Signal Review Committees replace informal oversight. Statistics enter the workflow to help identify and prioritize potential signals across a broader evidence base.

Balance defines this model. Human experts retain decision authority; analytics now support that authority in a substantive way. Governance also crosses a threshold, shifting from informal review to defined committees and repeatable processes. This is the groundwork on which enterprise-scale consistency is later built.

## Large pharmaceutical organizations: An analytics-enabled model

Large organizations manage high-volume, global datasets spanning many products and markets. Manual review cannot keep pace, so multilayered, analytics-driven programs combine internal data with major external sources and apply automated screening across stratified patient populations. Enterprise platforms supply the technology backbone; global governance frameworks maintain control across regions and functions.

Automation buys scalability, speed, and consistency. It also generates noise. High-volume screening surfaces large numbers of statistical associations, many of which carry no clinical meaning. The expert's role shifts accordingly rather than disappearing. At this scale, experts validate and interpret analytical outputs, filter noise, and ensure that every decision remains clinically sound and defensible under inspection.

## What stays constant: The critical success factors

The methodologies differ. The principles behind them do not. Four factors separate effective signal management programs from merely compliant ones.



**Medical judgment is fundamental**

Technology detects patterns and statistical associations. It cannot weigh biological plausibility, causality, confounding, or benefit-risk implications. Those judgments demand clinical expertise, and across every organizational size, expert judgment remains central to regulator-acceptable signal management.



**Governance ensures consistency**

Signal detection effectiveness rests on clearly defined structures, escalation pathways, and decision-making accountability. Governance formalizes as organizations grow, advancing from informal oversight to Signal Review Committees and ultimately to global multilayered frameworks. That formalization preserves consistency, transparency, and inspection readiness at scale.



**Technology enables; it does not replace**

Dashboards, statistical screening, and enterprise platforms exist to improve efficiency, scalability, and consistency. Position any of them as a substitute for clinical assessment and the accountability regulators expect begins to erode.



**Transparent documentation underpins trust**

A signal decision is only as strong as the record behind it. Clear, traceable documentation of what was reviewed, what was decided, and why, is essential to inspection readiness and to the credibility of the program as a whole.

## Human-in-the-loop: The operating model that holds at every scale

One finding matters more than any other. A human-in-the-loop model is the common thread linking the clinical-led biotech to the analytics-enabled global enterprise.

Small organizations apply the principle through direct medical review of individual cases. Large organizations apply it by using expert review to validate and interpret analytics-generated outputs. The mechanics differ; the intent is identical. Each combines the efficiency of technology with the accountability and rigor of expert clinical assessment.

Leadership oversight keeps that balance intact. As automation absorbs more of the screening burden, leadership ensures these tools strengthen efficiency and inspection readiness without weakening clinical accountability. A human-in-the-loop model absorbs growing data complexity while preserving regulator-trusted decisions. It is not a way station en route to full automation. It is the operating model itself.

## Building a fit-for-purpose program

Signal detection methodology tracks organizational scale, portfolio complexity, and safety data volume, and it stays grounded in consistent regulatory principles. Qualitative, physician-led assessment serves small organizations. Hybrid methods and formal governance serve mid-size organizations. Analytics-enabled, enterprise-scale frameworks serve large organizations. The failure mode is a mismatch: over-engineering a small program, or under-resourcing the governance of a large one.

The foundations do not change with scale. Expert medical judgment, strong governance and leadership oversight, transparent documentation, appropriate use of technology, and human-in-the-loop decision-making carry every program, regardless of size. Organizations that hold to these foundations while right-sizing their methodology maintain compliance, improve operational efficiency, and make scientifically sound, regulator-trusted safety decisions.

Safety data volumes will keep rising and analytical tools will keep advancing. The organizations that stay ahead scale their methods with intent and keep expert judgment at the center. Designing and operating signal management programs that fit an organization's scale, and that pair advanced analytics with deep clinical expertise, is an area where IQVIA brings extensive experience and a proven point of view.

Future AI innovations are expected to enhance signal management through predictive signal detection, intelligent signal prioritization, and greater automation of routine activities. While these capabilities can improve efficiency and support earlier identification of potential safety risks, expert medical judgment and a human-in-the-loop approach will remain essential to ensure scientific rigor, contextual interpretation, and patient-focused decision-making.



# About the authors



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Dr. Mohit Raizada is a seasoned pharmacovigilance and signal management professional with over 15 years of experience in drug safety, risk management, and safety analytics. As Director, Safety Reporting & Analytics at IQVIA, India, Dr. Raizada leads global signal management programs and partners with pharmaceutical and biotechnology organizations on safety strategy, regulatory compliance, and innovative approaches to signal detection and benefit-risk assessment. He is recognized for driving operational excellence, digital innovation, and data-driven decision-making to enhance patient safety outcomes.

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